

Animal rhythms

The world is full of repeating changes, or rhythms, that affect animals' lives. They include the 24-hour rhythm of day and night, the year-long rhythm of the seasons, and the rhythm of the tides. All over the world, animals keep in step with these changes, so that they can feed and breed when conditions are best. Animals do not have calendars, and they cannot plan ahead. They stay in step by sensing changes around them and by using built-in body clocks, which are kept ticking by hormones and by nerves.

Mass awakening

In North America, red-sided garter snakes hibernate together in burrows and crevices underground. When spring brings warmth and plentiful food, it is time to breed. The males emerge first. When a female appears, up to 100 males fight to be her partner, forming a writhing mass called a mating ball. The timing of this amorous frenzy varies according to the local climate. In the southern US, garter snakes can mate as early as February, but in Canada's cold climate, they may not emerge until June.



Breeding with the tides

The rhythm of the tides, controlled by the motion of the Sun and the Moon, dominates seashore life and even affects animals far out to sea. The time between high tides is normally 12.5 hours, but extra-large "spring" tides occur every two weeks.

Slender fish called California grunion use these tides to breed. When the spring tide is at its height, they wriggle their way onto beaches, where they mate and lay their eggs. A month later, during another spring tide, the eggs hatch, and the young grunion are swept out to sea.

